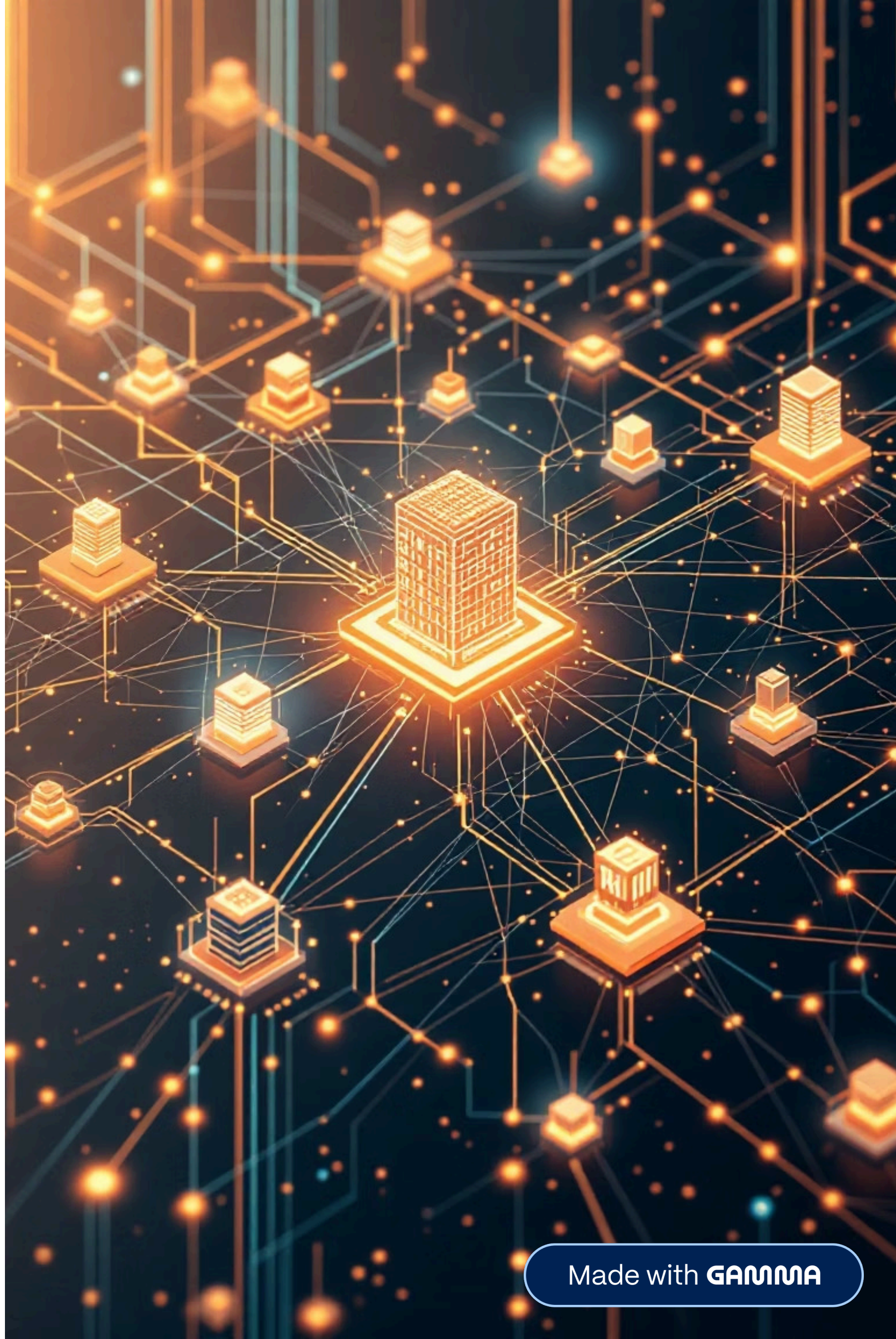
Fighting AI-Powered Threats with AI-Generated Defense

How synthetic attack generation is revolutionizing IoT threat detection for critical infrastructure

STEALTH STARTUP - RESEARCH PREVIEW

IOTSF WEBINAR



The Detection Gap: Why Traditional IDS Misses Sophisticated IoT Attacks

The 2025 threat landscape reveals an unprecedented escalation in AI-powered attacks targeting critical infrastructure and IoT networks. Traditional defenses are being systematically bypassed.

39M

Daily Attacks

Attack attempts hitting honeypot networks every 24 hours

71B

Infrastructure Hits

Attack attempts on critical infrastructure in 2024

400%

IoT Growth

Year-over-year increase in IoT-targeted attacks

78%

Bypass Rate

Traditional security measures evaded by AI-powered threats

Source: **Shieldworkz 2025 Threat Report** — 906 unique threat signatures targeting manufacturing sector alone

The Fundamental Problem with Traditional Detection

Three dominant approaches to intrusion detection are systematically failing against modern adversaries. Each methodology has critical blind spots that sophisticated attackers exploit.

Signature-Based

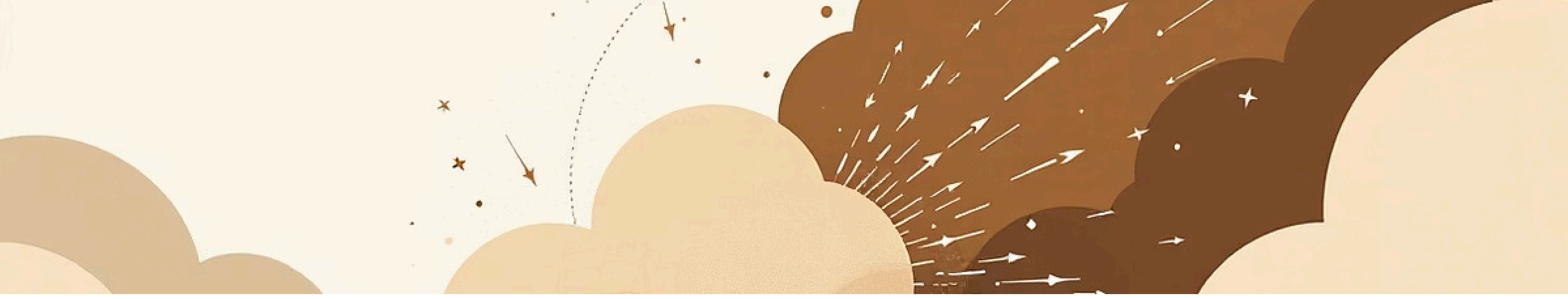
- Detects only known attack patterns
- Zero-day attacks remain invisible
- 76% of threats use polymorphic techniques
- Code rewritten per infection

Threshold Statistical

- Flags values outside normal range
- Sophisticated attacks stay within bounds
- High false positives overwhelm SOCs
- Easily evaded with rate-limiting

Traditional ML

- Trained only on known attacks
- Never sees evasion tactics
- Blind to traffic-mimicking attacks
- Limited attack diversity in data



The "Hard-Negative" Gap in IoT Security

Sophisticated attackers craft malicious traffic that is statistically indistinguishable from legitimate operations. Traditional detectors, never exposed to these patterns during training, classify them as benign.

Hard-Negative Definition: A malicious sample that appears statistically identical to benign traffic. Traditional detectors never encounter these during training, resulting in production detection failures.

APT41 Power Grid Campaigns

6-months loiter attacks using legitimate protocols and administrative traffic patterns to evade detection while establishing persistent access

Living-off-the-Land Techniques

Adversaries weaponize legitimate system tools and trusted binaries, making malicious activity indistinguishable from normal operations

Slow Exfiltration Attacks

Data theft matching backup traffic patterns, timing, and volume characteristics to blend seamlessly with authorized transfers

Visualizing Hard-Negative Examples

BEFORE: Traditional Detection

Packet Rate
▲
|
1000 | ██████ DDoS
|
100 |
|
10 | ●●●●●●●●
| ●●●●●●●●●●
Benign traffic
| ●●●●●●●●●
1 | ●●●●●●●
|
└──────────────
───────────────
— Data Volume
1KB 10KB 100KB 1MB

A simple decision boundary detects obvious attacks but ignores traffic within the "normal" range.

PROBLEM: Hard-Negatives Missed

Packet Rate
▲
|
1000 | ██████ DDoS
(✓)
|
100 |
|
10 | ●●●●●●●●
| ●●●●◆●●●●●● ←
APT41 ( )
| ●●●◆●●●●●● ← LoT
( )
1 | ●●●●●●●
| ◆ ← C2 beacon ( )
└──────────────
───────────────
— Data Volume

Sophisticated hard-negative attacks blend with benign traffic, evading traditional detection models.

AFTER: With Hard-Negative Training

Packet Rate
▲
|
1000 | ██████ DDoS
(✓)
|
100 |
| NEW TIGHTER
BOUNDARY
10 | ●●●●╱●●●●
| ●●●╱◆●●●●●● ←
APT41 (✓)
| ●●╱◆●●●●●● ← LoT
(✓)
1 | ●╱●●●●●
| ╱ ◆ ← C2 beacon (✓)
└──────────────
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— Data Volume

Hard-negative training refines boundaries, enabling detection of subtle anomalies within benign-looking traffic.

Our Approach: Generate Synthetic Hard-Negatives, Then Learn to Detect Them

Instead of waiting for attacks to emerge, we use AI to synthesize sophisticated threats before they appear in the wild—then train detection systems to recognize them or ***instead of building better mousetraps we engineer more evolved mice***!

Generate

Diffusion-TS creates synthetic attacks with benign statistical properties through trend and seasonality decomposition

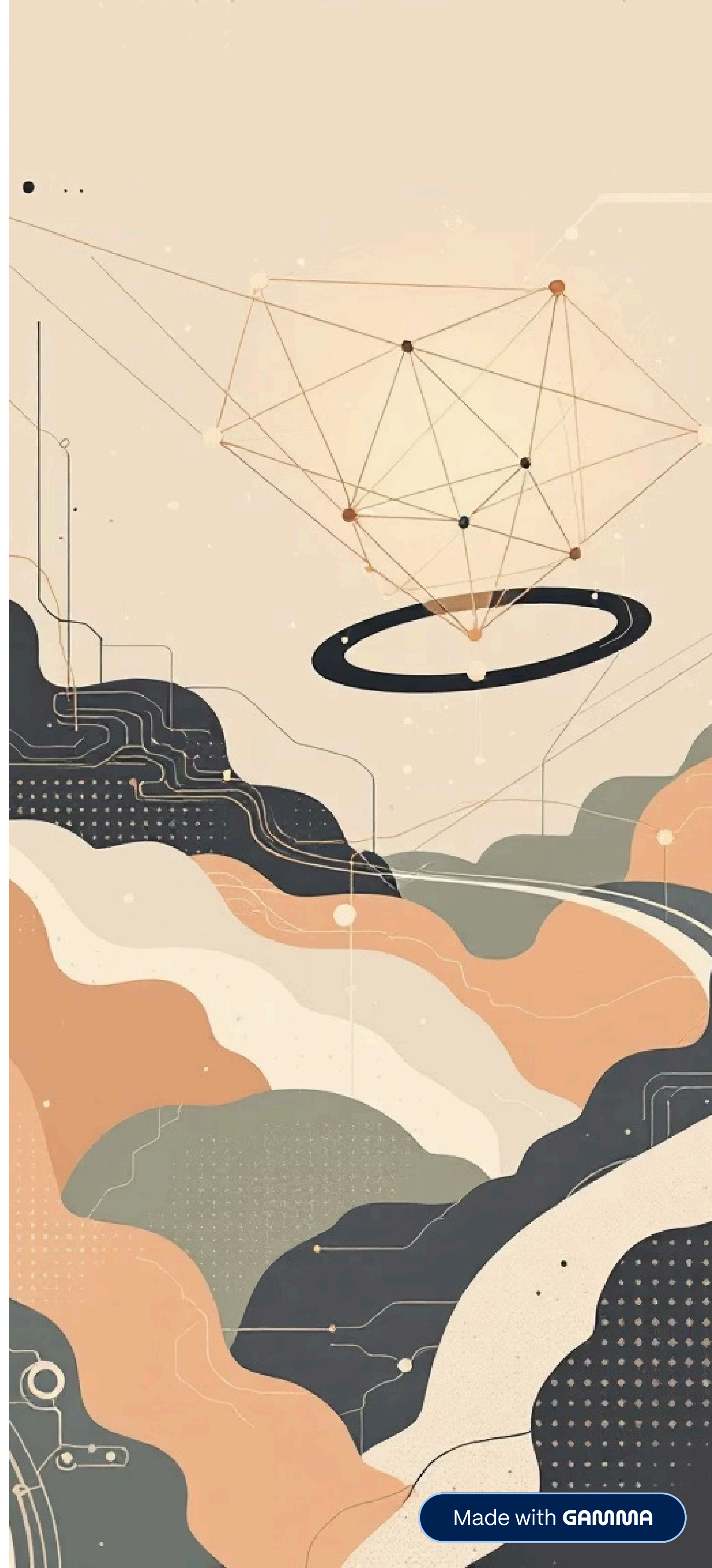
Constrain

GuidedDiffTime ensures protocol validity using hard constraints for packet rules and soft constraints for statistics

Detect

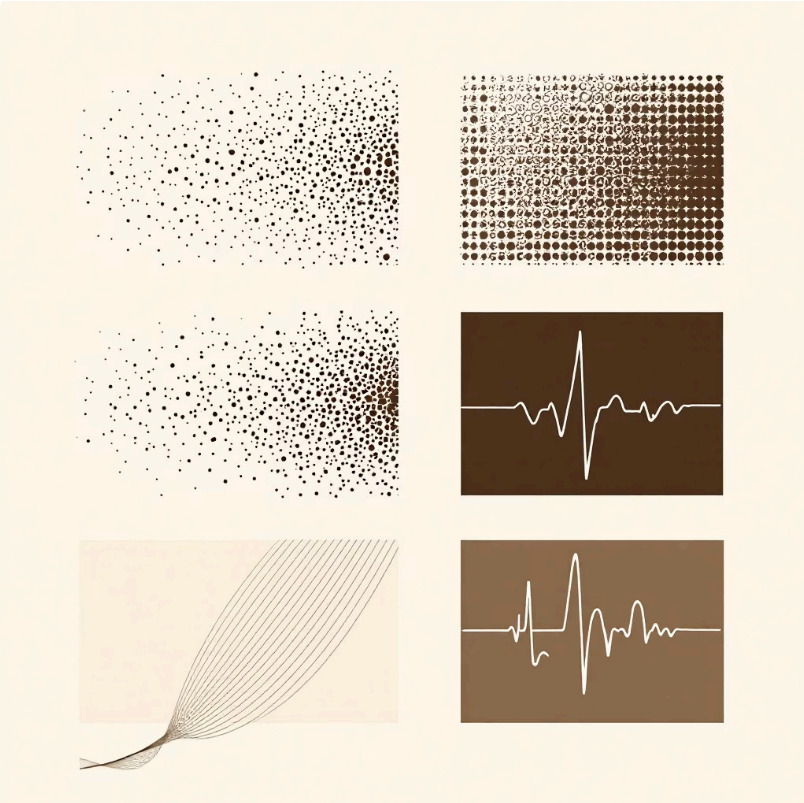
Moirai TSFM time-series foundation model pre-trained on 27B observations, fine-tuned on our hard-negatives

> We use AI to anticipate attacks that don't exist yet, then train our detector on them before they appear in the wild.



Diffusion-TS: Interpretable Time-Series Generation

Our generative backbone transforms random noise into realistic IoT traffic that maintains authentic statistical properties while embedding malicious patterns.



Key Innovations

- **Decomposition-Aware:** Separately models trend and seasonality components for authentic traffic patterns
- **Direct Reconstruction:** Predicts clean signal at each denoising step, not just noise reduction
- **Decomposed Output:** Generated traffic can be analyzed by trend/seasonality components ~ Interpretable
- **Guided Generation:** Targets specific statistical properties for hard-negative creation

01

Trend Component

Gradual directional changes matching legitimate device behavior over time

02

Seasonality Component

Periodic patterns reflecting regular polling intervals, backup schedules, and business hours

03

Combined Output

Synthetic traffic indistinguishable from benign baseline in statistical analysis

GuidedDiffTime: Making Attacks That Actually Work

Protocol-invalid attacks are useless—firewalls drop them immediately. We enforce constraints during generation to create threats that are simultaneously stealthy and valid.

Hard Constraints

Must satisfy for protocol validity

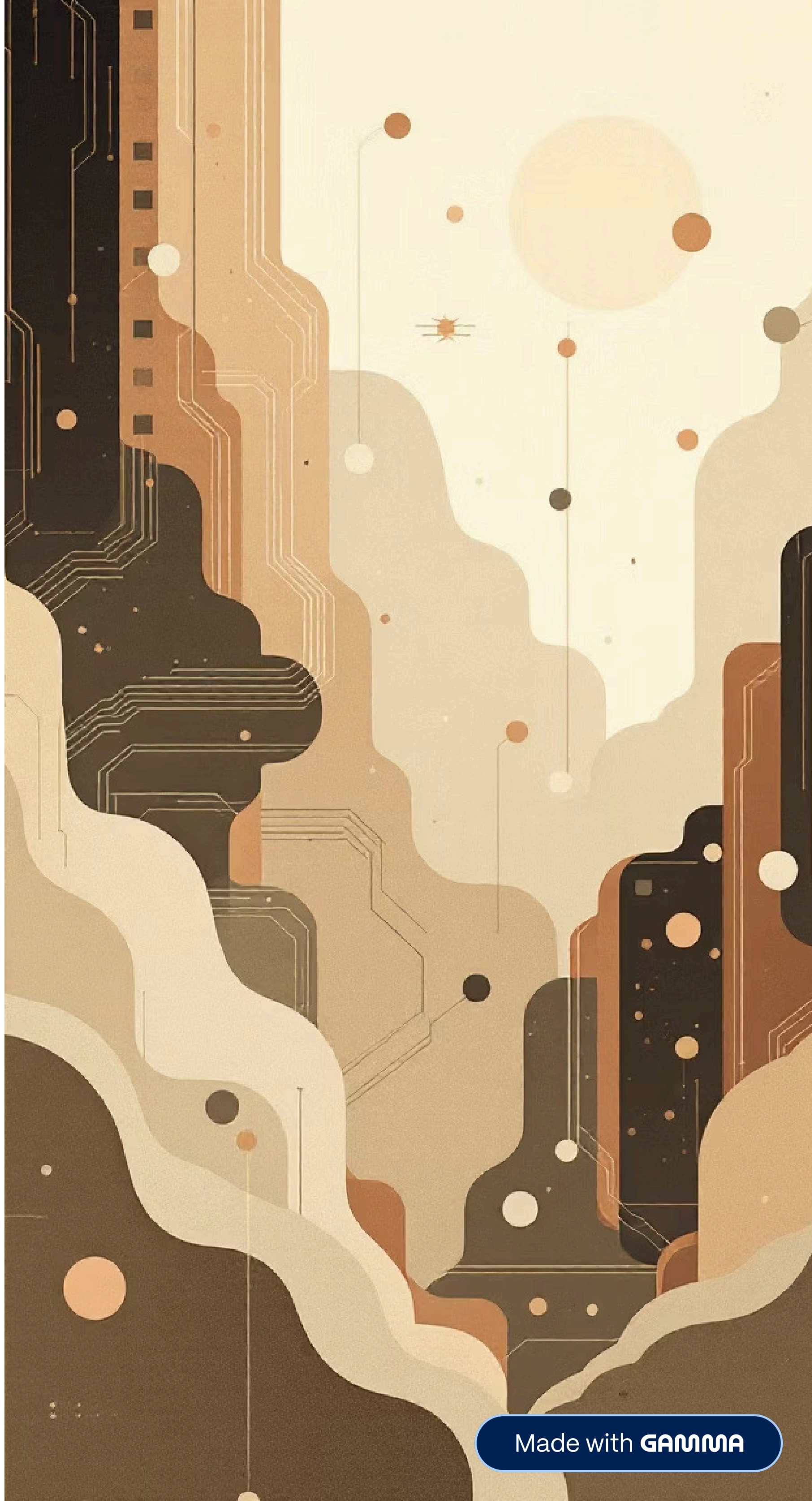
- Packet sizes within MTU limits
- Valid TCP flag sequences
- Non-negative timing values
- Protocol-specific rules (Modbus, MQTT, CoAP)

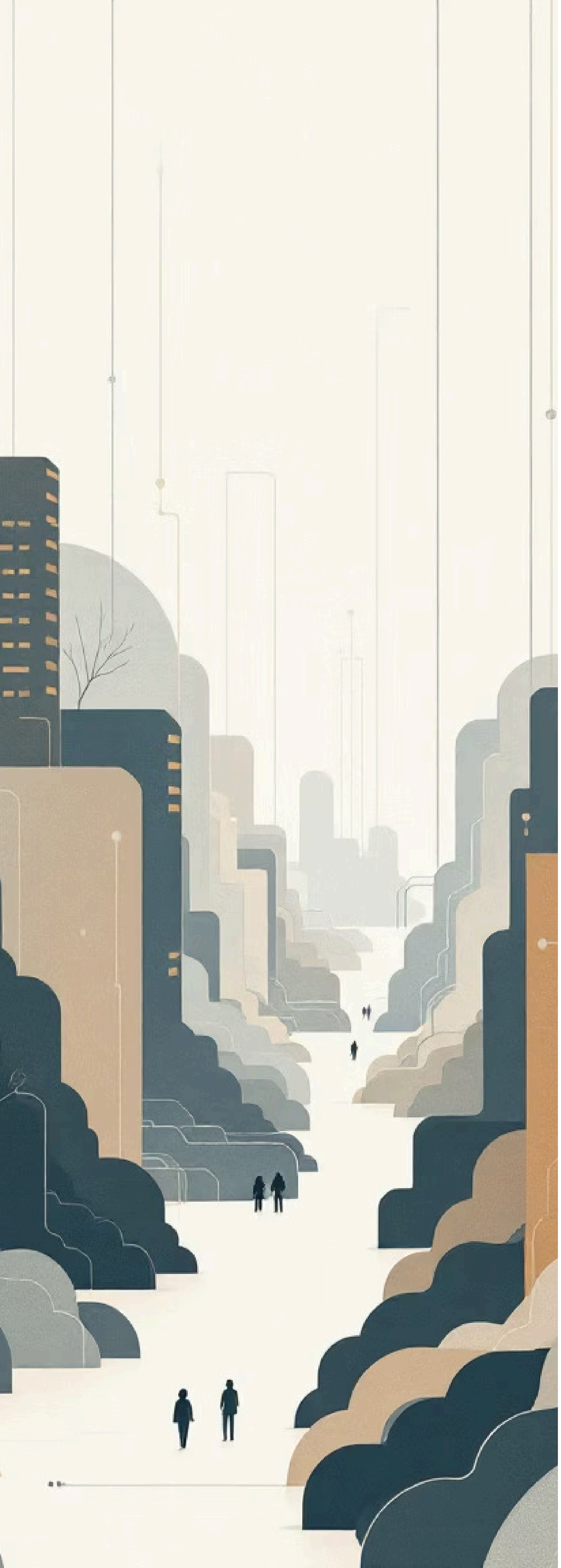
Soft Constraints

Statistical targets for stealth

- Mean within 5% of benign baseline
- Variance matching distribution
- Autocorrelation preservation
- Inter-arrival time alignment

Constraints are enforced **during generation**, not after. At each denoising step, we project the latent representation onto the valid manifold—ensuring every synthetic attack is both realistic and executable.





Moirai: Zero-Shot Anomaly Detection at Scale

We leverage a time-series foundation model pre-trained on 27 billion observations to achieve unprecedented detection capabilities across diverse IoT environments.

Foundation Model Architecture

Pre-trained on 27B observations across 9 domains including IT operations, energy systems, and cloud telemetry—understands universal temporal patterns

Any-Variate Attention

Single model handles devices with any number of sensors—from simple thermostats (1 variable) to complex turbines (50+ variables) without retraining

Zero-Shot Forecasting

Moirai predicts future time-series values with uncertainty estimates, even for datasets not seen during training.

Application to Anomaly Detection (Our Approach):

We use Moirai's probabilistic predictions to flag anomalies: - Predict expected traffic patterns - Compare observed values to predictions - Flag deviations outside confidence intervals as potential threats

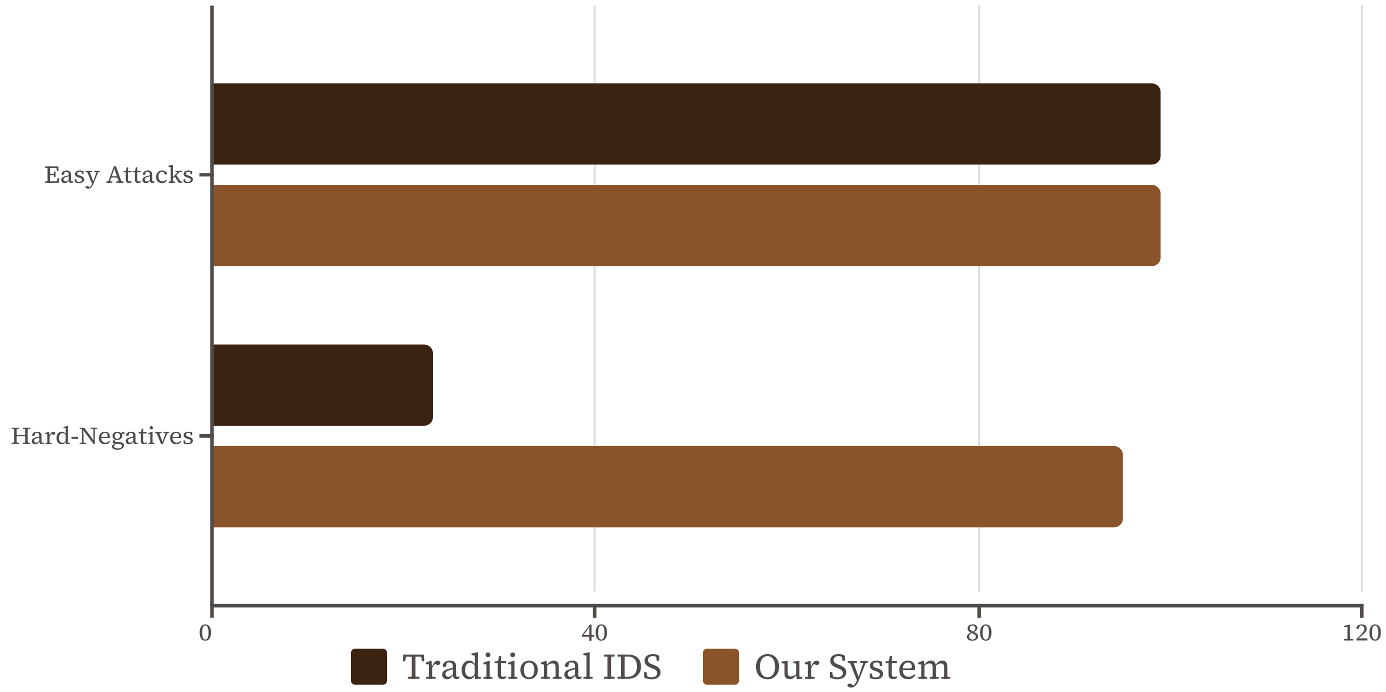
> The breakthrough: Fine-tuning on our synthetic hard-negatives teaches Moirai to detect subtle attack signatures that traditional ML never learned—because it never saw them in training data.

Validation on Real-World Data (work in progress)

Testing against CICIoT2023, the industry-standard benchmark with 105 real IoT devices and 33 attack types. (indicative chart*)

CICIoT2023 Dataset

- 105 real IoT devices
- 33 attack types, 7 categories
- 548 GB network traffic
- Industry-standard benchmark



Regulatory Context

Critical infrastructure operators face increasing regulatory pressure:

- **NIS2 Directive:** Requires "appropriate" security measures
- **EU AI Act:** Governs AI use in high-risk applications

Our approach supports compliance goals by:

- Improving threat detection capabilities
- Using synthetic training data (privacy-preserving)
- Providing explainable anomaly scores

Note: Specific compliance depends on implementation context

The Future of IoT Security is Adaptive

01

Traditional IDS Can't Detect Sophisticated Attacks

Threats designed to mimic normal traffic systematically evade signature-based and statistical detection

02

AI-Generated Hard-Negatives Fill the Training Gap

Synthetic attack generation creates the curriculum needed to teach detectors about advanced evasion techniques

03

Foundation Models Enable Better Detection

Pre-trained models once fine-tuned work across diverse IoT environments without device-specific retraining

04

Regulatory-Compliant and Production-Ready

Meets NIS2, EU AI Act, and GDPR requirements while delivering superior threat detection

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Early Adopter Program

We're forming a consortium for manufacturing, utilities, and oil & gas sectors. Validate against your specific threat data and shape the future of IoT defense.

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Next: Live Demo

In the next 10 minutes, you'll attempt to identify attacks in real traffic samples—then see how traditional IDS performs versus our system.

The threat landscape has fundamentally changed. The only viable response is a defense that evolves faster than the attack.